

## 8

# One-Way Repeated Measures ANOVA

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Does psychotherapy continue to work after a client stops seeing their therapist? Does exposing oneself to a person increase the chances that the person will like you? In this chapter, I'm going to answer both of these questions with a statistic known as the one-way repeated measures ANOVA (a.k.a., one-way within-subjects ANOVA), which is useful for testing the null hypothesis that two *or more* repeatedly measured means are equal. The means are considered repeatedly measured, because all of the participants are tested on the dependent variable two or more times. Because the one-way repeated measures ANOVA is a generalization of the paired samples *t*-test, there is no good reason for the paired samples *t*-test to be used anymore (just like the independent samples *t*-test). However, researchers traditionally use the paired samples *t*-test to test the difference between two within-subjects means, and the repeated measures ANOVA to test the difference between three or more within-subjects means. Consequently, this text includes material on the paired samples *t*-test (Chapter 6).

### One-way Repeated Measures ANOVA: Three Means

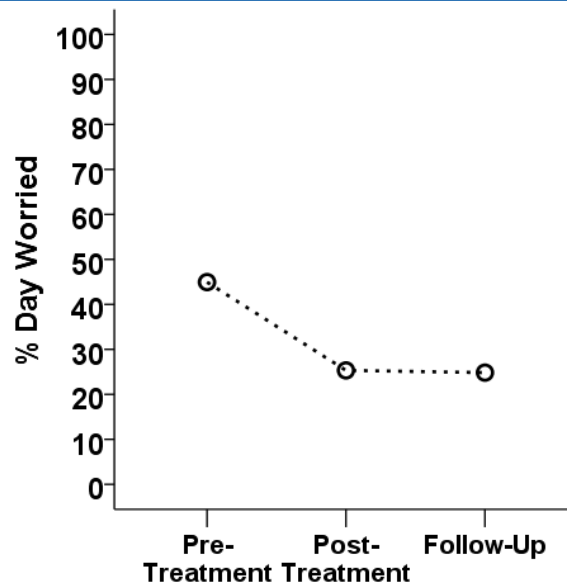
In the chapter relevant to the paired samples *t*-test, the Wetherell et al. (2003) CBT and anxiety study was included as part of the additional examples. Specifically, the null hypothesis that pre-test and post-test mean anxiety levels were equal was tested. The null hypothesis of equal means was rejected. Most researchers are not completely impressed with such evidence. That is, a key question researchers wonder about is whether those who receive treatment continue to experience the benefits of treatment, after they complete the treatment program. That is, if CBT really works, the clients should continue to apply the techniques they learn during treatment, once they stop seeing their therapist. Such a question should be asked by any researcher applying virtually any treatment or intervention program (e.g., goal-setting), not just psychotherapy. Consequently, good quality treatment/intervention studies often include a follow-up condition.

A follow-up condition is a period in which the participants who experienced the treatment are re-visited and measured on the same outcome/dependent variable(s) that were administered during the baseline and post-treatment conditions. Typically, follow-up conditions will take place several weeks, and often several months, after the last treatment session. I didn't mention this in the 'Difference Between Two Means' chapter (chapter 6), but Wetherell et al. (2003) included a follow-up condition in their investigation. That is, the clients were tested again on the measures six months after the last treatment session. Specifically, six months after the last treatment session, the clients responded to the following question, again: "On an average day over the past month, what percentage of the day did you feel worried?". I have displayed the three means associated with the three conditions in a line chart displayed in Figure C8.1. At least numerically, it looks like the CBT had a long-lasting benefit for the clients. Of course, the data need to be analyzed statistically.

In this case, the appropriate analysis to test the null hypothesis of three equal means (baseline, post-treatment, follow-up) associated with a within-subjects design is a repeated measures ANOVA, because the dependent variable was measured on an interval/ratio scale. It is known as a one-way analysis, because there is only one within-subjects independent variable: CBT treatment. The dependent variable is anxiety. As per the between-subjects ANOVA, I make the distinction between testing three within-subjects means and four or more within-subjects means, because the preferred follow-up multiple-comparison procedures differ ([Data File: Wetherell et al \(2003\) ideal](#)).

### C8.2

Figure C8.1. Plot of Wetherell et al (2003) Anxiety Means



### Hypotheses

Null Hypothesis ( $H_0$ ): The pre-treatment, post-treatment, and follow anxiety level means are equal across all three conditions.

Alternative Hypothesis ( $H_1$ ): The pre-treatment, post-treatment and follow-up anxiety level means are not all equal.

### One-way Within-Subjects ANOVA: Three Means: SPSS

In order to conduct a repeated measures ANOVA in SPSS, you can use the 'Repeated Measurements' menu option (**Watch Video 8.1: [Oneway Repeated Measures ANOVA in SPSS](#)**):

The first table outputted by SPSS is entitled 'Descriptive Statistics' and includes the means, standard deviations, and sample size. It can be observed in the 'Descriptive Statistics' table that the follow-up mean was numerically smaller ( $M = 24.83$ ) than the pre-treatment mean ( $M = 44.94$ ), as well as the post-treatment mean ( $M = 25.33$ ). Don't be fooled by the  $N = 18$ . There were a total of 18 participants in this within-subjects study (not  $3 * 18$ ).

Descriptive Statistics

|                | Mean  | Std. Deviation | N  |
|----------------|-------|----------------|----|
| pre_treatment  | 44.94 | 20.380         | 18 |
| post_treatment | 25.33 | 22.713         | 18 |
| follow_up      | 24.83 | 15.779         | 18 |

### C8.3

## CHAPTER 8: ONE-WAY WITHIN-SUBJECTS ANOVA

Next, a SPSS table entitled “Multivariate Tests” is included in the output, however, it can be ignored, in this case. The next SPSS table of interest is entitled “Mauchly’s Test of Sphericity”. I deal with the assumption of sphericity in detail in another section of this chapter below. For now, I will simply point out that Mauchly’s test of sphericity was not significant statistically ( $p = .292$ ). Consequently, we can proceed with the one-way within-subjects ANOVA in a straightforward manner.

Mauchly's Test of Sphericity<sup>a</sup>

Measure: MEASURE\_1

| Within Subjects Effect | Mauchly's W | Approx. Chi-Square | df | Sig. | Epsilon <sup>b</sup> |             |             |
|------------------------|-------------|--------------------|----|------|----------------------|-------------|-------------|
|                        |             |                    |    |      | Greenhouse-Geisser   | Huynh-Feldt | Lower-bound |
| time                   | .857        | 2.465              | 2  | .292 | .875                 | .967        | .500        |

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

- Design: Intercept  
Within Subjects Design: time
- May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

The SPSS table entitled ‘Tests of Within-Subjects Effects’ includes the test of the omnibus null hypothesis of equal anxiety means across time (pre-treatment, post-treatment, and follow-up). This is the key one-way repeated measures ANOVA result. Because Mauchly’s test of sphericity was satisfied, one can focus on the results in the rows labelled “Sphericity Assumed”. In this case, the null hypothesis of equal means was rejected,  $F(2, 34) = 14.59$ ,  $p < .001$ , partial  $\eta^2 = .462$ . Notice how SPSS refers to the effect size as partial eta-squared, rather than simply eta-squared (as per the between-subjects one-way ANOVA). I discuss the diction between  $\eta^2$  and partial  $\eta^2$  in further detail below.

Tests of Within-Subjects Effects

Measure: MEASURE\_1

| Source      |                    | Type III Sum of Squares | df     | Mean Square | F      | Sig. | Partial Eta Squared |
|-------------|--------------------|-------------------------|--------|-------------|--------|------|---------------------|
| time        | Sphericity Assumed | 4735.815                | 2      | 2367.907    | 14.593 | .000 | .462                |
|             | Greenhouse-Geisser | 4735.815                | 1.750  | 2706.015    | 14.593 | .000 | .462                |
|             | Huynh-Feldt        | 4735.815                | 1.935  | 2447.999    | 14.593 | .000 | .462                |
|             | Lower-bound        | 4735.815                | 1.000  | 4735.815    | 14.593 | .001 | .462                |
| Error(time) | Sphericity Assumed | 5516.852                | 34     | 162.260     |        |      |                     |
|             | Greenhouse-Geisser | 5516.852                | 29.752 | 185.429     |        |      |                     |
|             | Huynh-Feldt        | 5516.852                | 32.888 | 167.749     |        |      |                     |
|             | Lower-bound        | 5516.852                | 17.000 | 324.521     |        |      |                     |

Unfortunately, SPSS does not automatically report the correlations between the conditions in a within-subjects ANOVA. You have to perform the analysis separately (**Watch**

### C8.4

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**Video 8.2:** [Within-Subjects Correlations](#)). When I did, I obtained the following Pearson correlations:

|                |                     | Correlations  |                |           |
|----------------|---------------------|---------------|----------------|-----------|
|                |                     | pre_treatment | post_treatment | follow_up |
| pre_treatment  | Pearson Correlation | 1             | .754**         | .347      |
|                | Sig. (2-tailed)     |               | .000           | .158      |
|                | N                   | 18            | 18             | 18        |
| post_treatment | Pearson Correlation | .754**        | 1              | .650**    |
|                | Sig. (2-tailed)     | .000          |                | .004      |
|                | N                   | 18            | 18             | 18        |
| follow_up      | Pearson Correlation | .347          | .650**         | 1         |
|                | Sig. (2-tailed)     | .158          | .004           |           |
|                | N                   | 18            | 18             | 18        |

\*\* . Correlation is significant at the 0.01 level (2-tailed).

As can be seen in the SPSS table entitled 'Correlations', the Pearson correlation between anxiety at pre-treatment and post-treatment was positive and substantial at  $r = .75$ . However, the positive correlation between pre-treatment anxiety and follow-up anxiety was numerically lower at  $r = .35$ . Finally, the correlation between post-treatment and follow-up was somewhere in the middle at  $r = .65$ . This is the sort of pattern you can expect in practice: positive correlations which decrease in magnitude, at least numerically, as the time period between the two conditions increases.<sup>1</sup>

The positive correlations across conditions has the effect of reducing the size of the mean square error term in the one-way within-subjects ANOVA. Specifically, the 'Error(time)' Mean Square term in the SPSS table entitled 'Tests of Within-Subjects Effects' was equal to 162.26. Had there not been positive correlations between the three conditions, the Mean Square Error term would have been approximately 395.21 (based on some additional calculations I performed), which would have resulted in a much smaller  $F$ -value (i.e., 5.99). The positive correlations between repeatedly measured conditions is at the heart of the within-subjects design power advantage over the between-subjects design.

The SPSS table entitled 'Tests of Within-Subjects Contrasts' is interesting, but I deal with it in detail in the section relevant to 'trend analyses' later in the chapter.

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<sup>1</sup> Truth be told, Wetherell et al. (2003) did not report the correlations in their manuscript, unfortunately. Based on the results that they did report, I believe the correlations between the conditions were near zero, which is rather surprising.

## CHAPTER 8: ONE-WAY WITHIN-SUBJECTS ANOVA

Tests of Within-Subjects Contrasts

Measure: MEASURE\_1

| Source      | time      | Type III Sum of Squares | df | Mean Square | F      | Sig. |
|-------------|-----------|-------------------------|----|-------------|--------|------|
| time        | Linear    | 3640.111                | 1  | 3640.111    | 16.502 | .001 |
|             | Quadratic | 1095.704                | 1  | 1095.704    | 10.542 | .005 |
| Error(time) | Linear    | 3749.889                | 17 | 220.582     |        |      |
|             | Quadratic | 1766.963                | 17 | 103.939     |        |      |

Finally, and curiously, SPSS has outputted a table entitled 'Tests of Between-Subjects Effects'. Do not get confused here. There are no conventionally considered between-subjects effects in one-way repeated measures ANOVAs. What SPSS reports in this table are the results associated with testing the hypothesis that the overall mean collapsed across all three conditions (i.e., averaged) is equal to zero. In this case, the overall mean was equal to 31.7, based on  $(44.94 + 25.33 + 24.83) / 3 = 31.7$ . Thus, the difference between 31.7 and zero was found to be statistically significant,  $F(1,17) = 63.43$ ,  $p < .001$ . Why anyone would want to know the answer to such a question is beyond me. I suppose some possibly interesting questions could be posed, if one had a mix of positive and negative values in the data.

Tests of Between-Subjects Effects

Measure: MEASURE\_1

Transformed Variable: Average

| Source    | Type III Sum of Squares | df | Mean Square | F      | Sig. | Partial Eta Squared |
|-----------|-------------------------|----|-------------|--------|------|---------------------|
| Intercept | 54276.741               | 1  | 54276.741   | 63.431 | .000 | .789                |
| Error     | 14546.593               | 17 | 855.682     |        |      |                     |

In summary, the null hypothesis of equal anxiety means was rejected,  $F(2, 34) = 14.59$ ,  $p < .001$ . Furthermore, the pattern of means suggested that the anxiety levels decreased from pre-treatment to post-treatment, and from post-treatment to follow-up. However, as per the one-way between-subjects ANOVA, a statistically significant one-way repeated measures ANOVA really just tells us that we have more work to do. Specifically, we have to discover precisely which means differ from which means. You may recall that we use multiple comparison procedures to deal with this issue. First, however, I describe the conventional measure of an effect size in the context of a repeated measures ANOVA.

### One-way Within-Subjects ANOVA: Effect Size

Eta squared ( $\eta^2$ ) cannot be used as a measure of effect size in a one-way within-subjects ANOVA, because there is an additional term within the model. The additional term represents the positive correlations between the conditions. Consequently, researchers tend to report partial eta squared, which is based on the following formula:

### C8.6

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$$\text{partial } \eta^2 = \frac{SS_{\text{Effect}}}{SS_{\text{Effect}} + SS_{\text{Error}}} \quad (1)$$

$SS_{\text{Effect}}$  refers to the within-subjects factor effect. That is, the effect due to the independent variable (e.g., CBT treatment). In this example,  $SS_{\text{Effect}}$  was equal to 4735.82. Instead of  $SS_{\text{Within}}$ , SPSS uses the term  $SS_{\text{Error}}$  in the repeated measures ANOVA case. In this example,  $SS_{\text{Error}}$  was equal to 5516.85. Consequently,

$$\text{partial } \eta^2 = \frac{4735.82}{4735.82 + 5516.85} = .462$$

It is inappropriate to interpret partial  $\eta^2$  as the proportion of variance in the dependent variable that can be accounted for by the independent variable. Instead, in the one-way repeated measures case, partial  $\eta^2$  can be interpreted as the proportion of variance in the dependent variable accounted for by the independent variable, excluding the variance accounted for by the correlations between the conditions.<sup>2</sup> Thus, in the Wetherell et al. (2003) example, 46.2% of variance in anxiety levels was accounted for by the CBT treatment, excluding the shared variance between the anxiety scores across time.

### One-way Within-Subjects ANOVA: Three Means – Multiple Comparisons

When I discussed the null hypothesis associated with the one-way between-subjects ANOVA with three means, I recommended the protected Fisher's LSD as the most optimal multiple comparison procedure, because it has been shown to be the most powerful, while also keeping the familywise error rate at .05 (Hayter, 1986). Although the analytical and simulation work that has established the utility of protected Fisher's LSD as a multiple comparison procedure has been based on the between-subjects ANOVA, I am very confident that the implications extend to the one-way repeated measures ANOVA. I look forward to someone confirming (or shaking) my confidence on this issue. Consequently, in the psychotherapy and anxiety levels study, I have performed three Fisher's LSD multiple comparison tests to follow-up the statistically significant omnibus repeated measures ANOVA. SPSS has produced a table entitled 'Pairwise Comparisons' which includes the Fisher's LSD results (**Watch Video 8.3: Repeated Measures ANOVA Protected Fisher's LSD in SPSS**):

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<sup>2</sup> If you conduct a one-way between-subjects ANOVA in GLM within SPSS, SPSS will curiously label the  $\eta^2$  value as partial  $\eta^2$  when it is in fact  $\eta^2$  (Lakens, 2013). Be careful. Partial  $\eta^2$  exists only in the context of one-way within-subjects ANOVA and factorial ANOVAs (see Chapters 9, 10, and 11).

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### Pairwise Comparisons

Measure: MEASURE\_1

| (I) time | (J) time | Mean Difference (I-J) | Std. Error | Sig. <sup>b</sup> | 95% Confidence Interval for Difference <sup>b</sup> |             |
|----------|----------|-----------------------|------------|-------------------|-----------------------------------------------------|-------------|
|          |          |                       |            |                   | Lower Bound                                         | Upper Bound |
| 1        | 2        | 19.611 <sup>*</sup>   | 3.601      | .000              | 12.013                                              | 27.209      |
|          | 3        | 20.111 <sup>*</sup>   | 4.951      | .001              | 9.666                                               | 30.556      |
| 2        | 1        | -19.611 <sup>*</sup>  | 3.601      | .000              | -27.209                                             | -12.013     |
|          | 3        | .500                  | 4.075      | .904              | -8.098                                              | 9.098       |
| 3        | 1        | -20.111 <sup>*</sup>  | 4.951      | .001              | -30.556                                             | -9.666      |
|          | 2        | -.500                 | 4.075      | .904              | -9.098                                              | 8.098       |

Based on estimated marginal means

\*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Least Significant Difference (equivalent to no adjustments).

As can be seen in the SPSS table entitled 'Pairwise Comparisons', SPSS does not include the variable names in the left column to specify the comparison, unfortunately. Instead, SPSS uses the numbers 1, 2, and 3 to refer to the time conditions. Thus, 1 and 2, in this case, corresponds to the pre-treatment and post-treatment comparison. The mean difference was equal to 19.61, standard error was equal to 3.60, and  $p = .000$ , or  $p < .001$  to be more appropriate. In order to obtain the  $t$ -values, divide the mean difference by the standard error. Thus, for the pre-treatment and the post-treatment mean comparison,  $t = 5.45$ ,  $p < .001$ . Additionally, the pre-treatment and follow-up mean comparison was also significant statistically,  $t = 4.06$ ,  $p = .001$ . However, the post-treatment and follow-up mean comparison was not statistically significant,  $t = 0.12$ ,  $p = .904$ . The fact that the participants did not evidence a statistically significant change in anxiety after post-treatment suggests that the psychotherapy continued to be effective. Had the anxiety levels increased statistically significantly after post-treatment, it may have been cause for concern.

It will be noted that that the  $t$ -values obtained from Fisher's LSD in this case are identical to those that would be obtained from conducting a series of paired-samples  $t$ -tests. This suggests that SPSS does not use a pooled error term for the comparisons. This is a bit of a surprise, considering SPSS does use a pooled error term for Fisher's LSD in the one-way between-subjects ANOVA case. Ideally, a pooled error term would be used here (**Watch Video 8.4: Fisher's LSD vs Paired  $t$ -test**).

### One-Way Within-Subjects ANOVA: Assumptions

The one-way repeated measures ANOVA is a parametric statistical analysis with a number of assumptions. The assumptions are identical to the between-subjects ANOVA with the exception that the one-way repeated measures ANOVA does not assume homogeneity of variance. Instead, it assumes a similar assumption known as sphericity.

### C8.8



## CHAPTER 8: ONE-WAY WITHIN-SUBJECTS ANOVA

### 1: Random sampling

All cases in the population must have an equal chance of being selected into the sample. In practice, it is entirely unrealistic to run a study such that every person (or case) in the population to which you wish to infer your results has an equal chance of being selected into your sample. Consequently, in practice, this first assumption is virtually always violated. The extent to which the violation of this assumption affects the accuracy of statistical results is anyone's guess. Ultimately, there is not much that can be done about it. The show must go on, as they say.

In the Wetherell et al. (2003) psychotherapy study, the participants were recruited from a particular treatment center, consequently, the assumption of random sampling was violated. Such a sample is known as a **convenience sample**.

### 2: Independence of observations

Independence of observations implies that the participants in the investigation have not influenced each other with respect to the dependent variable(s) of interest. Typically, it is easy to satisfy this assumption, if the participants complete the tasks, tests, or questionnaires independently. Of course, in a within-subjects design, the data are dependent across conditions, because the same participants are measured repeatedly. This is not a violation of the independence of observations assumptions. The assumption applies across participants within a condition, not across conditions.

In the Wetherell et al. (2003) psychotherapy and anxiety study, it is likely the case that the participants did not influence each other's anxiety levels, as the psychotherapy sessions were experienced individually.

### 3: Dependent variable measured on an interval/ratio scale

Theoretically, the one-way repeated measures ANOVA assumes that the dependent variable has been measured on an interval/ratio scale. However, a substantial amount of simulation research has found that parametric statistics, such as the Pearson correlation and the *t*-test, can yield accurate results when the dependent variable has been measured on an ordinal scale with at least 5-points (O'Brien & Homer, 1987; Rasmussen, 1989). If the dependent variable has been measured on a scale with 4-points or less, however, then a non-parametric statistic such as the Friedman's ANOVA should be considered, for example, to test the difference between two or more within-subject means.

In the Wetherell et al. (2003) psychotherapy and anxiety study, the dependent variable was measured on a ratio scale (0 to 100), consequently, a paired samples *t*-test should be considered appropriate.

### 4: Normally distributed data

This assumption is only true in theory. In practice, the data associated with a repeated measures ANOVA do not need to be normally distributed. Based on my review of the simulation research, the repeated measures ANOVA is very robust to violations of the normality assumption up to skew of  $|2.0|$  and kurtosis of  $|9.0|$ , if not even more robust than that. For example, Berkovits, Hancock, and Nevitt (2000) found that the regular repeated measures ANOVA  $F$ -test kept the type I error rate at .039, even when skew was 3.0 and kurtosis was 21.0 and the sample size was at least 10. Thus, as the type I error rate was relatively close to the expected value of .05, the repeated measures ANOVA should be considered very robust to violations of the theoretical assumption of normality.

I'll note that while much of the normality research with the dependent samples  $t$ -test has focused on difference scores only, the normality research on the within-subjects ANOVA tends to look only at the raw score distributions, not the difference scores. I'm not entirely sure of the implications of this, however, I suspect that whether one looks at the raw scores or the difference scores does not make a substantial difference, in most cases.

Based on the Wetherell et al. (2003) data, the pre-treatment (skew = .68; kurtosis = 1.37), post-treatment (skew = 1.19; kurtosis = .47), and follow-up (skew = .89; kurtosis = .67) anxiety scores were associated with skew and kurtosis levels less than  $|2.0|$  and  $|9.0|$ . Thus, the data should be considered sufficiently normal for the purposes of conducting a normal theory estimation repeated measures ANOVA (and Fisher's LSD multiple comparisons).

### 5: Sphericity

Sphericity is essentially a homogeneity of variance assumption. In that sense, the one-way repeated measures ANOVA is similar to the between-subjects ANOVA which also assumes homogeneity of variance. However, with respect to the one-within subjects ANOVA, the homogeneity of variance assumption incorporates information about the covariance between the conditions, in addition to the variances. Further details about the precise nature of sphericity are provided in the Advanced Topics section of this chapter.

If the assumption of sphericity has been violated, there is nothing to worry about. The Huynh-Feldt adjustment is a remarkable and widely available method to protect oneself against violations of the assumption of sphericity. For example, the Huynh-Feldt adjusted  $F$ -test will keep the type I error rate at .057, even when sphericity is as bad as epsilon = .48 and the sample size is as small as 10 (Berkovits, Hancock, & Nevitt, 2000). Furthermore, with a sample size of 30 or more, the Huynh-Feldt adjusted  $F$ -test will keep the type I error rate at approximately .05, even when the data are extremely skewed (3.0) kurtotic (21.0), and sphericity is as large as epsilon .48. When sphericity is totally absent, the Huynh-Feldt adjusted  $F$ -test will work just as well as the regular  $F$ -test. So, why not use the Huynh-Feldt adjusted  $F$ -test in all cases? Good question. We probably should.

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The Greenhouse-Geisser adjusted  $F$ -test is fairly consistently more conservative than the Huynh-Feldt adjusted  $F$ -test, so it should probably be avoided, in most cases. The exception would possibly be when the data are very non-normally distributed (skew  $> 2.0$  and kurtosis  $> 9.0$ ), the sphericity assumption has been violated substantially (epsilon  $.57$  or less), and the sample size is less than 30. In such cases, the Greenhouse-Geisser adjusted  $F$ -test is probably the best option with respect to keeping the type I error rate close to  $.05$ .

Based on the Wetherell et al. (2003) data, the Mauchly's test of sphericity was not significant,  $\chi^2(2) = 2.47$ ,  $p = .292$ , Huynh-Feldt epsilon =  $.967$ . Thus, the assumption of sphericity was satisfied. Furthermore, epsilon was estimated at  $.967$ , which is very close to  $1.0$  (that's a good thing) (**Watch Video 8.5:** [Mauchly's Test of Sphericity in SPSS](#)).

### One-Way Repeated Measures ANOVA: Error Bars

Although I'm not a fan of including error bars in graphs which include means, many researchers tend to do so. In the context of a within-subjects design, the calculation of the standard error of the mean is more complicated, because there is typically a positive correlation between the dependent variable scores across the conditions. In the context of analyses such as the paired-samples  $t$ -test and the repeated measures ANOVA, the positive correlations between the conditions reduces the amount of error within the data, which also reduces the size of the standard error of the difference between the means. Consequently, it is arguably inappropriate to calculate the standard error of the means based on the typical formula ( $SD/\sqrt{N}$ ). Instead, an alternative formulation must be employed which takes into account the positive correlations between the conditions. In practical terms, the alternative formulation must adjust the size of the standard deviations. That is, they should be smaller than what is estimated from the raw data.

A number of procedures have been suggested to accomplish the adjustments required to obtain valid within-subjects confidence intervals for means (Cousineau & O'Brien, 2014; Loftus & Masson, 1994; Morey, 2008). I find the procedures lack intuitive appeal. Additionally, they tend to involve a large number of steps.

As described in the chapter devoted to tests of the difference between two means, I described a simpler method, based on subtracting the covariance from the variances. However, in the context of an ANOVA, the procedure can be performed more efficiently by square rooting the  $MS_W$  from the within-subjects ANOVA table. The square root of the  $MS_W$  corresponds to the average adjusted standard deviations (**Watch Video 8.6:** [Within-Subjects Adjusted Standard Deviations](#)). As there is only one  $MS_W$  in a one-way within-subjects ANOVA, there is only one adjusted  $SD$ . Thus, each condition will be given the same adjusted  $SD$  (i.e., pooled adjusted  $SD$ ). In the Wetherell et al. one-way within-subject ANOVA example, the  $MS_W$  was estimated at  $162.26$ . Therefore, the pooled  $SD_{adj} = \sqrt{162.26} = 12.738$ . Once the pooled adjusted  $SD$  is obtained, the following two steps can be performed:

#### C8.11

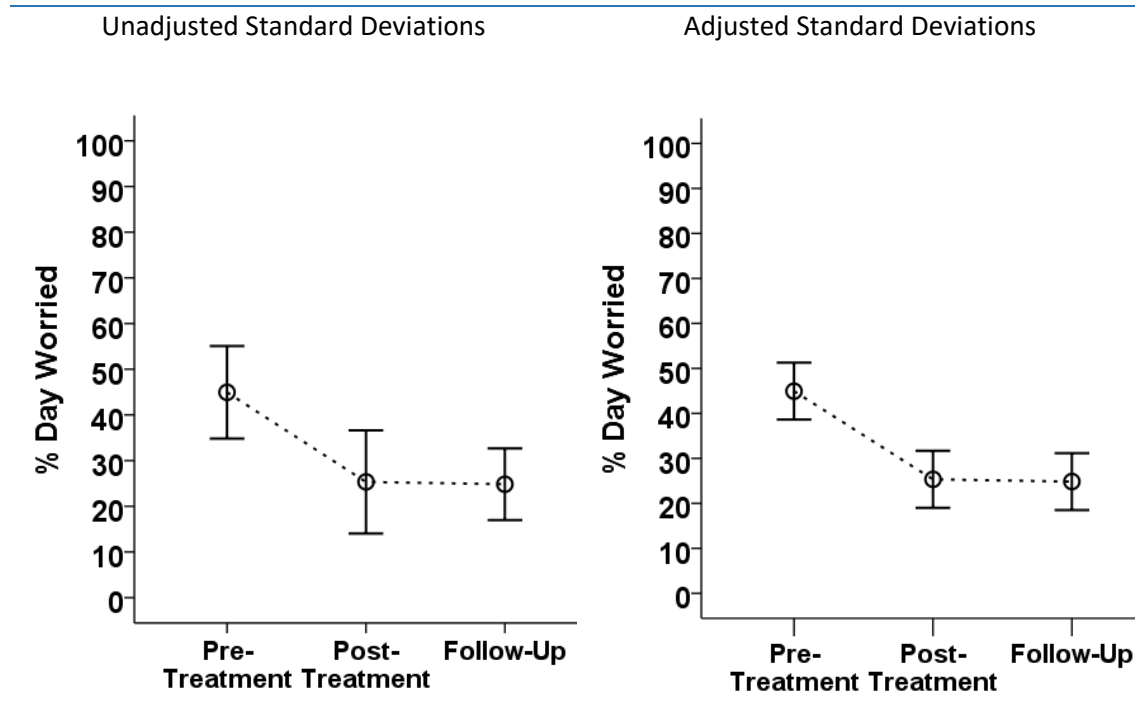
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- 1: Convert the original raw scores into z-scores.
- 2: Transform the z-scores to have the original mean values and the adjusted standard deviations (like the procedure for a *T*-score, for example).

That's it. You can then use the adjusted/transformed scores to create error bars with accurate 95% confidence intervals in a program like SPSS (**Watch Video 8.7:** [Within-Subjects Adjusted Confidence Intervals](#)). I'll note that if the assumption of sphericity is violated, and you interpret the Huynh-Feldt adjusted *F*-value, you would be best to square root the Huynh-Feldt  $MS_W$  term, in order to obtain the pooled adjusted SD.

Figure C8.2 represents a line chart with adjusted 95% confidence intervals for the means associated with the one-way within-subjects ANOVA on the Wetherell et al. data.

*Figure C8.2: Plot of Wetherell et al (2003) Means and 95% Confidence Intervals*



### One-way Repeated Measures ANOVA: Four or More Means

The mere exposure effect suggests that humans like something more the more they get exposed to it. Such an effect has been demonstrated for a large number of objects (Zajonc, 2001). Phetersen and Horai (1976) were interested to determine whether the mere exposure effect extended to human faces. That is, do people rate particular faces as more likable, if those people get exposed to the particular faces more often, in comparison to faces to which they were exposed fewer times?

## CHAPTER 8: ONE-WAY WITHIN-SUBJECTS ANOVA

### Hypotheses

Null Hypothesis ( $H_0$ ): Mean ratings of likability for faces will be equal across all levels of exposure.

Alternative Hypothesis ( $H_1$ ): Two or more mean ratings of likability for faces will be unequal.

To address the hypotheses, Phetersen and Horai (1976) exposed 100 adult participants to 20 different images of people with variable levels of attractiveness. Each image was presented for 2 seconds with a half second interval between each displayed image. Importantly, Phetersen et al. (1976) presented some of the images with varying levels of frequency: 1, 2, 5, 10 and 25 times. The images were counterbalanced with respect to frequency exposure to the participants, and the order of presentation of the images was randomized. At the end of the exposures, the participants had to rate the degree to which they liked an image on a scale from 1 to 7, with a higher score indicative of more liking ([Data File: exposure likability](#)).

| Descriptive Statistics |        |                   |     |
|------------------------|--------|-------------------|-----|
|                        | Mean   | Std.<br>Deviation | N   |
| exposure1              | 4.3660 | 1.23103           | 100 |
| exposure2              | 4.3629 | 1.14329           | 100 |
| exposure5              | 4.4932 | 1.34277           | 100 |
| exposure10             | 4.4470 | 1.38282           | 100 |
| exposure25             | 4.8022 | 1.04492           | 100 |

I have simulated data to correspond very closely to the results reported in Phetersen and Horai (1976). The means and standard deviations associated with the Phetersen et al. (1976) study are reported in the SPSS table entitled 'Descriptive Statistics'. It can be seen that the mean level of liking was equal to 4.37 for images that were exposed only once (exposure1). By contrast, the mean level of liking was equal to 4.80 for images that were exposed 25 times (exposure25). Furthermore, across all levels of exposure, there was a suggestion for an increasing trend in the means. The numerical differences in the means and the possible trend are interesting, however, it is necessary to test these numerical effects for statistical significance, which can be done with a repeated measures ANOVA.

In order to conduct a one-way repeated measures ANOVA with more than three means, one can utilize the 'Repeated Measurements' menu option (**Watch Video 8.8: [Oneway Within-Subjects ANOVA with 4+ Levels in SPSS](#)**). In this case, the selected multiple comparison procedure must be Bonferonni, rather than LSD, because there are more than three means in the analysis. As can be seen in the output table entitled 'Machly's Test of Sphericity', the assumption of sphericity has been satisfied, as Mauchly's statistic was not significant statistically,  $\chi^2(9) = 14.91$ ,  $p = .093$ . Had Mauchly's been significant (i.e.,  $p < .05$ ), it would have

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## CHAPTER 8: ONE-WAY WITHIN-SUBJECTS ANOVA

been necessary to consult the Huynh-Feldt adjusted values associated with the repeated measures ANOVA. However, as it was not significant, it is appropriate to consult the sphericity assumed results.

**Mauchly's Test of Sphericity<sup>a</sup>**

Measure: MEASURE\_1

| Within Subjects Effect | Mauchly's W | Approx. Chi-Square | df | Sig. | Epsilon <sup>b</sup> |             |             |
|------------------------|-------------|--------------------|----|------|----------------------|-------------|-------------|
|                        |             |                    |    |      | Greenhouse-Geisser   | Huynh-Feldt | Lower-bound |
| exposure               | .858        | 14.914             | 9  | .093 | .930                 | .971        | .250        |

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept

Within Subjects Design: exposure

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

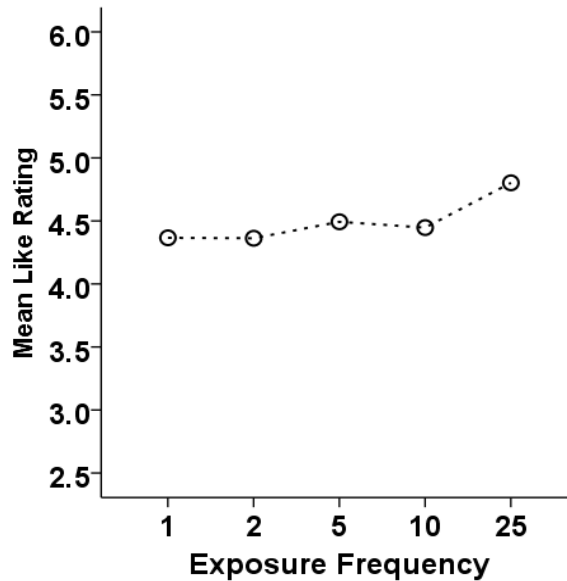
Next, SPSS produced a table entitled 'Tests of Within-Subjects Effects', which includes the results associated with the one-way repeated measures ANOVA. It can be seen that the null hypothesis of no difference between the means has been rejected,  $F(4, 396) = 6.85$ ,  $p < .001$ , partial  $\eta^2 = .065$ . As the direction of the means was such that greater amounts of exposure was associated with greater liking, the mere exposure effect may be suggested to affect peoples' liking of images of other people. Such an effect can be appreciated by an examination of Figure C8.3.

**Tests of Within-Subjects Effects**

Measure: MEASURE\_1

| Source          |                    | Type III Sum of Squares | df      | Mean Square | F     | Sig. | Partial Eta Squared |
|-----------------|--------------------|-------------------------|---------|-------------|-------|------|---------------------|
| exposure        | Sphericity Assumed | 13.080                  | 4       | 3.270       | 6.854 | .000 | .065                |
|                 | Greenhouse-Geisser | 13.080                  | 3.721   | 3.515       | 6.854 | .000 | .065                |
|                 | Huynh-Feldt        | 13.080                  | 3.884   | 3.367       | 6.854 | .000 | .065                |
|                 | Lower-bound        | 13.080                  | 1.000   | 13.080      | 6.854 | .010 | .065                |
| Error(exposure) | Sphericity Assumed | 188.937                 | 396     | .477        |       |      |                     |
|                 | Greenhouse-Geisser | 188.937                 | 368.366 | .513        |       |      |                     |
|                 | Huynh-Feldt        | 188.937                 | 384.539 | .491        |       |      |                     |
|                 | Lower-bound        | 188.937                 | 99.000  | 1.908       |       |      |                     |

*Figure C8.3. Plot of Likability Means*



Typically, researchers follow-up a statistically significant ANOVA with a series of mean comparisons to help identify which mean differences are statistically significantly different from each other. In the case where three means are analysed, Fisher's LSD is the most powerful and justifiable approach to keeping the per contrast error rate at .05 for both the between-subjects and the repeated measures one-way ANOVA.

However, for cases with more than three means, one must employ an alternative strategy to help ensure the familywise error rate does not exceed .05 appreciably. Unfortunately, there are very other few within-subjects multiple comparison procedures that have been developed, thus far. In SPSS, the only options are Bonferonni and Sidak, excluding Fisher's LSD, which is only appropriate for designs that include three means. The Sidak procedure is slightly less conservative than the Bonferonni procedure. It will maintain the familywise error rate at .05, when the selected comparisons are independent from each other, which I believe would be true in most applied cases, but perhaps not all cases. Realistically, most researchers are not in a position to determine whether all of their multiple comparisons are independent from each other. Consequently, in the absence of any other alternatives, the Bonferonni method is recommended as the best multiple comparison procedure in the one-way repeated measures ANOVA case with four or more means (unless a trend analysis is a viable option, as discussed further below).

In order to conduct a series of Bonferonni corrected multiple comparisons for a one-way repeated measures ANOVA in SPSS, be sure to select the 'Options' button within the repeated measures ANOVA window (do not selected the 'Post-Hoc' button). (**Watch Video 8.9: Oneway Within-Subjects Bonferroni in SPSS**).

In the current example, SPSS produced a table entitled 'Pairwise Comparisons' which includes all of the Bonferroni correct pairwise comparison results. It can be seen that the only

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comparisons that were statistically significant ( $p < .05$ ) were those that included the condition with the 25 exposures. Thus, from this perspective, 25 exposures were required for the mere exposure effect to take place in a statistically significant manner.

### Pairwise Comparisons

Measure: MEASURE\_1

| (I) exposure | (J) exposure | Mean Difference (I-J) | Std. Error | Sig. <sup>b</sup> | 95% Confidence Interval for Difference <sup>b</sup> |             |
|--------------|--------------|-----------------------|------------|-------------------|-----------------------------------------------------|-------------|
|              |              |                       |            |                   | Lower Bound                                         | Upper Bound |
| 1            | 2            | .003                  | .104       | 1.000             | -.295                                               | .301        |
|              | 3            | -.127                 | .100       | 1.000             | -.413                                               | .159        |
|              | 4            | -.081                 | .110       | 1.000             | -.398                                               | .236        |
|              | 5            | -.436*                | .102       | .000              | -.729                                               | -.144       |
| 2            | 1            | -.003                 | .104       | 1.000             | -.301                                               | .295        |
|              | 3            | -.130                 | .087       | 1.000             | -.380                                               | .119        |
|              | 4            | -.084                 | .102       | 1.000             | -.376                                               | .208        |
|              | 5            | -.439*                | .091       | .000              | -.699                                               | -.179       |
| 3            | 1            | .127                  | .100       | 1.000             | -.159                                               | .413        |
|              | 2            | .130                  | .087       | 1.000             | -.119                                               | .380        |
|              | 4            | .046                  | .101       | 1.000             | -.244                                               | .337        |
|              | 5            | -.309*                | .094       | .014              | -.579                                               | -.039       |
| 4            | 1            | .081                  | .110       | 1.000             | -.236                                               | .398        |
|              | 2            | .084                  | .102       | 1.000             | -.208                                               | .376        |
|              | 3            | -.046                 | .101       | 1.000             | -.337                                               | .244        |
|              | 5            | -.355*                | .084       | .001              | -.596                                               | -.114       |
| 5            | 1            | .436*                 | .102       | .000              | .144                                                | .729        |
|              | 2            | .439*                 | .091       | .000              | .179                                                | .699        |
|              | 3            | .309*                 | .094       | .014              | .039                                                | .579        |
|              | 4            | .355*                 | .084       | .001              | .114                                                | .596        |

Based on estimated marginal means

\*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Bonferroni.

The Bonferroni correction is a single-step multiple comparison procedure. Therefore, it is unnecessary to first observe a statistically significant ANOVA. In fact, consulting and interpreting the ANOVA in the analysis reduces the power of the analysis (so avoid the temptation to look at ANOVA table!).

In many cases, I would be dissatisfied with ending a within-subjects investigation with the conclusion that only a handful of comparisons with a single condition evidenced a statistically significant effect. In the current example, it may be appropriate to make such a conclusion, as the pattern of means was such that there was no obvious increase in liking across all of the exposure conditions. However, in other cases, there will be an obvious numerical pattern of increasing (or decreasing means), but only a handful of multiple comparisons may be observed to be statistically significant. In such cases, a much more powerful approach to



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analyzing the data would to perform a trend analysis, as I demonstrate in the Advanced Topics section of the chapter based on slightly modified Phetersen and Horai (1976) exposure effect data.

## Advanced Topics

## Assumption of Sphericity

Unlike the between-subjects ANOVA, the within-subjects ANOVA does not assume homogeneity of variance. Instead, the within-subjects ANOVA assumes something called sphericity. Sphericity is a more complicated version of the homogeneity of variance assumption. The main difference is that sphericity takes into consideration the variances associated with the conditions, as well as the associations between the conditions. By associations, I mean correlations, or covariances more precisely. Consequently, when the variances and the covariances are equal across all conditions (within sampling fluctuations), the assumption of sphericity will be satisfied.

Typically, the assumption of sphericity is tested with Mauchly's test of sphericity. Mauchly's test of sphericity essentially tests the hypothesis that the denominator portions of the paired samples  $t$ -tests (see formula below) across all possible mean comparisons associated with the conditions in the design are equal (within sampling fluctuations). Recall that the denominator portion of the dependent samples  $t$ -test is also known as the standard error of the difference between two within-subjects means.

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{N} + \frac{S_2^2}{N} - \frac{2 \text{cov}_{12}}{N}}} \quad (2)$$

Therefore, in the case of a repeated measures ANOVA with three conditions (i.e., means), it would be implied that there are three possible pairwise comparisons (Mean 1 versus Mean 2; Mean 1 versus Mean 3; and Mean 2 versus Mean 3). Consequently, there would also be three standard error of the differences between two within-subjects means that need to be the same, within sampling fluctuations, in order for the assumption of sphericity to be satisfied. Sphericity is estimated with a statistical value known as epsilon ( $\epsilon$ ). The maximum value  $\epsilon$  can take is 1.0, which is indicative of perfect sphericity (a good thing). Smaller  $\epsilon$  values are indicative of greater differences in the variances/covariances across the condition comparisons (i.e., less likely to satisfy the assumption of sphericity).

In practice, the assumption of sphericity is rarely satisfied. There are two reasons for this. First, many within-subjects investigations incorporate some sort of treatment or independent variable to which the participants are exposed. However, in most cases, there are substantial individual differences in the degree to which the participants respond to the treatment. Consequently, the post-treatment condition variance is almost always larger than the pre-treatment condition variance. Secondly, the correlation (and covariance) between conditions tend to vary in magnitude, because time decreases similarity. For example, if you measure a group of peoples' anxiety at baseline, then again one month later, and then again

three months later, you can bet money that the correlation between anxiety at baseline and one month later will be larger than the correlation between anxiety at baseline and three months later. The reason is that, over time, people change (e.g., circumstances, people in their lives, employment, etc.). Consequently, more time facilitates more non-systematic change, which reduces correlations between scores across time.

Finally, I'll note that there is no possibility to test the assumption of sphericity when you only have two conditions in your analysis, as there is only one standard error of the difference between two means. Consequently, the assumption of sphericity does not apply in the paired samples *t*-test case, nor does it apply repeated measures ANOVAs with only two levels associated with the independent variable (**Watch Video 8.10:** [What is sphericity exactly?](#)).

What is a large Epsilon Value in the Context of Sphericity?  
(forthcoming)

### Trend Analysis

The most powerful approach to the evaluation of a hypothesized effect in a within-subjects design is to perform a trend analysis. A trend analysis is not a follow-up or multiple comparison procedure, consequently, there is no need to observe a statistically significant ANOVA. Thus, a researcher can proceed straight to a trend analysis. A trend analysis tests the hypothesis that the means are associated with a particular pattern in a statistically significant way. The two most common patterns that are tested are linear and quadratic. A statistically significant linear trend implies that statistically significant amount of the variance in the pattern of the means is consistent with linear contrast weightings. The straight line can either be upward or downward in direction. Loosely speaking, a linear trend is tested by testing a correlation between the observed means and set of weightings that are consistent with linear trend. In the context of the mere exposure study, a positive linear trend was arguably hypothesized: higher levels of exposure would be associated with higher levels of liking.

In order to demonstrate the utility of a trend analysis, I have modified the mere exposure effect data used in the previous example ([Data File: exposure likability modified](#)). A one-way repeated measures ANOVA conducted on the modified data yielded a statistically significant omnibus effect (**Watch Video 8.11:** [Trend Analysis in SPSS](#)). Thus, the null hypothesis of equal mean liking ratings can be rejected,  $F(4, 396) = 33.64$ ,  $p < .001$ , partial  $\eta^2 = .254$ . However, the follow-up Bonferroni adjusted multiple comparisons yielded a number of inter-condition comparisons that were not statistically significant, as can be seen in the SPSS table entitled 'Pairwise Comparisons'. Such inconsistency in the multiple comparisons makes for a difficult story to describe in a sample manner.

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### Pairwise Comparisons

Measure: MEASURE\_1

| (I) exposure | (J) exposure | Mean Difference (I-J) | Std. Error | Sig. <sup>b</sup> | 95% Confidence Interval for Difference <sup>b</sup> |             |
|--------------|--------------|-----------------------|------------|-------------------|-----------------------------------------------------|-------------|
|              |              |                       |            |                   | Lower Bound                                         | Upper Bound |
| 1            | 2            | -.147                 | .104       | 1.000             | -.445                                               | .151        |
|              | 3            | -.277                 | .100       | .065              | -.563                                               | .009        |
|              | 4            | -.431*                | .110       | .002              | -.748                                               | -.115       |
|              | 5            | -1.036*               | .102       | .000              | -1.328                                              | -.743       |
| 2            | 1            | .147                  | .104       | 1.000             | -.151                                               | .445        |
|              | 3            | -.130                 | .087       | 1.000             | -.380                                               | .119        |
|              | 4            | -.285                 | .102       | .061              | -.576                                               | .007        |
|              | 5            | -.889*                | .091       | .000              | -1.149                                              | -.629       |
| 3            | 1            | .277                  | .100       | .065              | -.009                                               | .563        |
|              | 2            | .130                  | .087       | 1.000             | -.119                                               | .380        |
|              | 4            | -.154                 | .101       | 1.000             | -.445                                               | .136        |
|              | 5            | -.759*                | .094       | .000              | -1.029                                              | -.488       |
| 4            | 1            | .431*                 | .110       | .002              | .115                                                | .748        |
|              | 2            | .285                  | .102       | .061              | -.007                                               | .576        |
|              | 3            | .154                  | .101       | 1.000             | -.136                                               | .445        |
|              | 5            | -.604*                | .084       | .000              | -.845                                               | -.363       |
| 5            | 1            | 1.036*                | .102       | .000              | .743                                                | 1.328       |
|              | 2            | .889*                 | .091       | .000              | .629                                                | 1.149       |
|              | 3            | .759*                 | .094       | .000              | .488                                                | 1.029       |
|              | 4            | .604*                 | .084       | .000              | .363                                                | .845        |

Based on estimated marginal means

\*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Bonferroni.

However, an examination of the plot of means in Figure C8.4 seems to reveal a pretty straightforward story: greater levels of exposure were associated with greater levels of liking. The trend analysis confirmed the simple story: there was a statistically significant linear (increasing) trend in the means,  $F(1, 99) = 97.35$ ,  $p < .001$ , partial  $\eta^2 = .496$ . A statistical trend analysis should always be conducted in combination with a visual depiction of the means in a graph. In this case, there was also a statistically significant quadratic effect,  $F(1, 99) = 13.74$ ,  $p < .001$ , partial  $\eta^2 = .122$ . A quadratic effect suggests that the effect was not completely linear. A quadratic effect implies that there is one “bend” in the pattern of the effect. In this example data, the “bend” occurred from exposure 10 to exposure 25. Thus, a statistically significant greater exposure effect was observed between 10 and 25 exposures, in comparison to the other exposure comparisons.

### C8.20

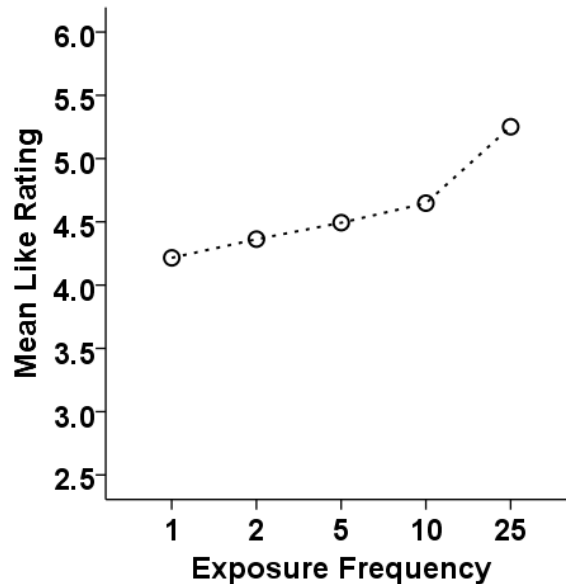
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Tests of Within-Subjects Contrasts

Measure: MEASURE\_1

| Source          | exposure  | Type III Sum of Squares | df | Mean Square | F      | Sig. | Partial Eta Squared |
|-----------------|-----------|-------------------------|----|-------------|--------|------|---------------------|
| exposure        | Linear    | 55.514                  | 1  | 55.514      | 97.354 | .000 | .496                |
|                 | Quadratic | 6.297                   | 1  | 6.297       | 13.738 | .000 | .122                |
|                 | Cubic     | 2.174                   | 1  | 2.174       | 4.679  | .033 | .045                |
|                 | Order 4   | .212                    | 1  | .212        | .511   | .476 | .005                |
| Error(exposure) | Linear    | 56.453                  | 99 | .570        |        |      |                     |
|                 | Quadratic | 45.379                  | 99 | .458        |        |      |                     |
|                 | Cubic     | 45.986                  | 99 | .465        |        |      |                     |
|                 | Order 4   | 41.114                  | 99 | .415        |        |      |                     |

Figure C8.4. Plot of Likability Means - Modified Data



### MANOVA as Alternative to Repeated Measures ANOVA

Some recommend using MANOVA (multivariate analysis of variance) in the place of a repeated measures ANOVA, when the assumption of sphericity has been violated (e.g., Vasey & Thayer, 1987). Although there may be some relatively rare circumstances under which the MANOVA would be a clearly better option over the repeated measures ANOVA, typically, using the Huynh-Feldt adjusted  $F$ -test will be the best option. Also, my impression is that the repeated measures ANOVA will test the hypothesis that researchers have in mind, whereas the repeated measures MANOVA will not quite do so. Consequently, I don't see any compelling reason to use repeated measures MANOVA over the repeated measures ANOVA, when the sphericity

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assumption has been violated, or even if the MANOVA is associated with more power, in some cases.

### One-way Repeated Measures ANOVA: Randomized Designs

The Wetherell et al. (2003) anxiety study had a clear order associated with the levels in the within-subjects factor: pre-test, post-test, and follow-up. Similarly, the Phetersen and Horai (1976) mere exposure study had a clear order associated with the levels in the within-subjects factor: 1, 2, 5, 10, and 25 exposures. However, not all within-subjects designs have an order to the levels in the within-subjects factor. Instead, the levels associated with the within-subjects design can be essentially arbitrary. It is common to observe such a within-subjects design in proper experiments. That is, where the administration of the conditions to the cases have been randomized.

For example, Friman, McPherson, Warzak, and Evans (1993) were interested in the social acceptance of thumb sucking behavior among 1<sup>st</sup> grade children. To investigate the issue, Friman et al. (1993) had 40 1<sup>st</sup> grade children view four pictures. Two of the pictures were of a 1<sup>st</sup> grade boy (unknown to the 40 children). In one of the pictures of the boy, the boy was sucking his thumb. In the other picture of the boy, the boy was not sucking his thumb. The other two pictures were of a 1<sup>st</sup> grade girl (again, unknown to the 40 children). In one of the pictures of the girl, the girl was sucking her thumb. In the other picture of the girl, the girl was not sucking her thumb. After viewing all four pictures, each of the 40 children were asked, individually, various questions pertaining to how much they liked the child. As Friman et al. (1993) wanted to counteract any possible carry-over effects, they randomized the order which the pictures were shown to the 40 children. I tested the null hypothesis that the mean perceived likability of the four pictures of the children were equal. ([Data File: thumb sucking acceptance](#)) ([Watch Video 8.12: Randomized Within-Subjects ANOVA Example in SPSS](#))

As can be seen in the SPSS table entitled 'Descriptive Statistics', the means were at least different numerically. For example, the 'girl thumb sucking' photo was associated with a mean of 17.25. By contrast, the 'girl not thumb sucking' photo was associated with a mean of 22.85.

**Descriptive Statistics**

|                  | Mean    | Std.<br>Deviation | N  |
|------------------|---------|-------------------|----|
| boy_sucking      | 17.3500 | 5.65000           | 40 |
| boy_not_sucking  | 20.1300 | 5.84000           | 40 |
| girl_sucking     | 17.2500 | 5.23000           | 40 |
| girl_not_sucking | 22.8500 | 9.12000           | 40 |

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Next, it can be seen in the SPSS table entitled ‘Tests of Within-Subjects Effects’ that the null hypothesis of equal means was rejected,  $F(3, 117) = 11.92$ ,  $p < .001$ , partial  $\eta^2 = .234$ . Finally, the multiple comparison results revealed that all of the ‘thumb sucking’ versus ‘not thumb sucking’ mean comparisons were statistically significant. Thus, the 1<sup>st</sup> grade children, on average, rated photos of other children sucking their thumb as less likable than photos of the same children not sucking their thumb.

### Pairwise Comparisons

Measure: MEASURE\_1

| (I) picture_type | (J) picture_type | Mean Difference (I-J) | Std. Error | Sig. <sup>b</sup> | 95% Confidence Interval for Difference <sup>b</sup> |             |
|------------------|------------------|-----------------------|------------|-------------------|-----------------------------------------------------|-------------|
|                  |                  |                       |            |                   | Lower Bound                                         | Upper Bound |
| 1                | 2                | -2.780*               | .900       | .022              | -5.281                                              | -.279       |
|                  | 3                | .100                  | .887       | 1.000             | -2.367                                              | 2.567       |
|                  | 4                | -5.500*               | 1.240      | .000              | -8.947                                              | -2.053      |
| 2                | 1                | 2.780*                | .900       | .022              | .279                                                | 5.281       |
|                  | 3                | 2.880*                | .835       | .008              | .560                                                | 5.200       |
|                  | 4                | -2.720                | 1.307      | .264              | -6.352                                              | .912        |
| 3                | 1                | -.100                 | .887       | 1.000             | -2.567                                              | 2.367       |
|                  | 2                | -2.880*               | .835       | .008              | -5.200                                              | -.560       |
|                  | 4                | -5.600*               | 1.263      | .000              | -9.110                                              | -2.090      |
| 4                | 1                | 5.500*                | 1.240      | .000              | 2.053                                               | 8.947       |
|                  | 2                | 2.720                 | 1.307      | .264              | -.912                                               | 6.352       |
|                  | 3                | 5.600*                | 1.263      | .000              | 2.090                                               | 9.110       |

Based on estimated marginal means

\*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Bonferroni.

In summary, the reason I have demonstrated this example of a repeated measures ANOVA is to ensure that you are aware that not all within-subject independent variables are associated with some sort of order. The conditions can be measured on a nominal scale.

### Understanding the Between-Subjects Effect in Repeated Measures ANOVA

As described in the Foundations section of this chapter, there is a between-subjects effect associated with the repeated measures ANOVA. However, the between-subjects effect, in this context, has nothing to do with possible mean differences between groups. Instead, the between-subjects effect is a test of the null hypothesis that the grand mean is zero. The grand mean is the mean of the means across the conditions. In the overwhelmingly vast majority of cases, the grand mean will not be zero. Additionally, it will be entirely uninteresting to test and/or report such a result. For illustration purposes, I have simulated hypothetical repeated

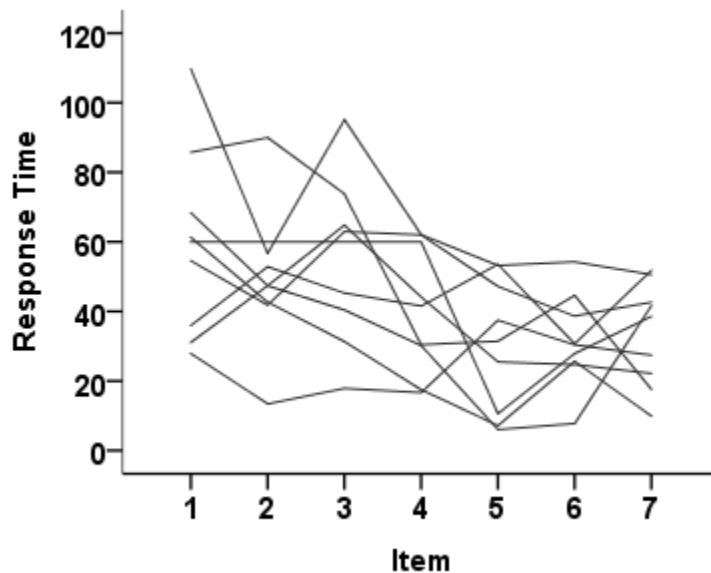
### C8.23

measures ANOVA data that are associated with the total absence of any between-subjects effect variance. Correspondingly, the grand mean associated with these simulated data is exactly zero ([Data File: grand\\_mean\\_zero](#)) ([Watch Video 8.13: Testing the Within-Subjects Grand Mean is Zero in SPSS](#)).

### Spaghetti Plot

For some studies, you might want to present the profile of scores across the levels of the within-subjects factor for all individual cases, rather than the means associated with the whole sample. Such plots are known as spaghetti plots, because they can look very messy, especially if the pattern in the case values is not strong and there are a lot of cases included in the spaghetti plot. Although messy, spaghetti plots can provide some interesting information about the consistency of the trend across all cases. For the purposes of demonstration, I created in SPSS ([Data File: anagrams\\_10](#)) a spaghetti plot for nine cases who completed the Anagram Persistence Test, which measures the amount of time people spend completing essentially impossible anagrams ([Watch this Video](#)). Typically, a spaghetti plot will only look good when the dependent variable has been measured on ratio/interval scale. In my experience, it will not look good for Likert data.

*Figure C8.5. Spaghetti Plot – APT-Difficult Response Time Data*





### Practice Questions

The following practice questions were developed for you to get experience with testing null hypotheses associated with three or more related (within-subjects) means. They are based on real studies and the results correspond closely to the results reported by the studies, so you may be interested to learn the results in their own right. Before you tackle the practice questions, I encourage you to watch me describe the steps I recommend following, when testing null hypotheses for research scenarios with three or more related (within-subjects means). The steps essentially summarize the key information I covered in this chapter. (**Watch Video 8.14: [Oneway Within-Subjects ANOVA - Steps](#)**). Once you have knowledge of the steps, apply the steps to helping uncover the research question statistically. Then, check the answers you obtained with the ones I obtained, as I describe in each practice question's corresponding video.

#### *1: Does Cognitive Behavioral Therapy Help People With Anxiety?*

You may recall in the Foundation section of this chapter that I covered a study relevant to anxiety levels and cognitive behavioral therapy (i.e., Wetherell, et al., 2003). You likely remember the general results, but not the specific results. For example, what was the  $F$ -value? What was the Mauchly's test of sphericity  $p$ -value? Consequently, given your familiarity with the Wetherell et al. (2003) study, as a first exercise, use the steps I recommend to test the following null hypothesis with the Wetherell et al. (2003) data (**Data File: [Wetherell et al ideal](#)**): the pre-treatment, post-treatment and follow-up means were equal. If you need help with a particular step, look into the chapter for further information. Ultimately, give yourself a chance to analyze the data with your stats program. Then, watch the following video to check your results (**Watch Video 8.P1: [Oneway Within-Subjects ANOVA - CBT and Anxiety \(Practice 1\)](#)**).

#### *2: Does winning the lottery increase happiness?*

Many people dream about experiencing a substantial financial windfall. However, does winning a large amount of money really make a difference in a person's overall level of happiness? Brickman, Coates, and Janoff-Bulman (1978) found 22 major lottery winners and interviewed them. One of the questions they asked was to rate their overall level of happiness on a six-point scale (0 not at all, 5 = very much). They asked the question three times: once in the context of their happiness in the past, their happiness at this stage in their life, and their expected future happiness. Test the null hypothesis that the happiness levels were equal across all three time periods (**Data File: [win lottery happiness](#)**) (**Watch Video 8.P2: [Oneway Within-Subjects ANOVA - Lottery Win \(Practice 2\)](#)**).

#### *3: Does not winning the lottery increase happiness?*

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Brickman, Coates, and Janoff-Bulman (1978) also found 22 people who had not won a major lottery and interviewed them, too, with the same sorts of questions. Importantly, one of the questions they asked was to rate their overall level of happiness on a six-point scale (0 not at all, 5 = very much). As per the lottery winners, they asked the non-winners the question three times: once in the context of their happiness in the past, their happiness at this stage in their life, and their expected future happiness. Test the null hypothesis that the happiness levels were equal across all three time periods (**Data File: no\_win\_lottery\_happiness**) (**Watch Video 8.P3: [Oneway Within-Subjects ANOVA - Lottery No Win \(Practice 3\)](#)**).

### *4: Does the value of hedonism change with age?*

Personal values are an important attribute of a person's psychological make-up. Hedonism is the value of sensual self-indulgence. Vecchione et al. (2016) examined the stability of hedonism across young adulthood age. The value of hedonism was measured (Likert-scale; greater than 5-point scale) at three time periods, separated by 4 years (ages 20, 24, and 28). Test the null hypothesis that the value of hedonism was equal across all three time periods (**Data File: hedonism\_across\_age**) (**Watch Video 8.P4: [Oneway Within-Subjects ANOVA - Hedonism \(Practice 4\)](#)**).

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